

Report1

Assignment 1

Data Analytics

E0-259

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Implementation summary for the assignment: -

To make sense out of data visual representation helps. Plotting graph of the gives a better idea about what would be the most appropriate distribution of network into clusters which are intensely related to each other. To sort graphs into closely related communities. To have sensible and useful communities different algorithms are used to obtain them. Girvan Newman and Louvain are two such algorithms.

Girvan Newman Algorithm - Communities are sorted by evaluating edge betweenness. Edge betweenness of each edge is calculated by considering each node a root one at a time and assigning a weight to the edge's dependent on the shortest path to the root from every node. This is done iteratively and then added and normalised. The edge with maximum edge betweenness is removed and the new graph again goes through iteration until there is no edge is left or a stopping criterion is met.

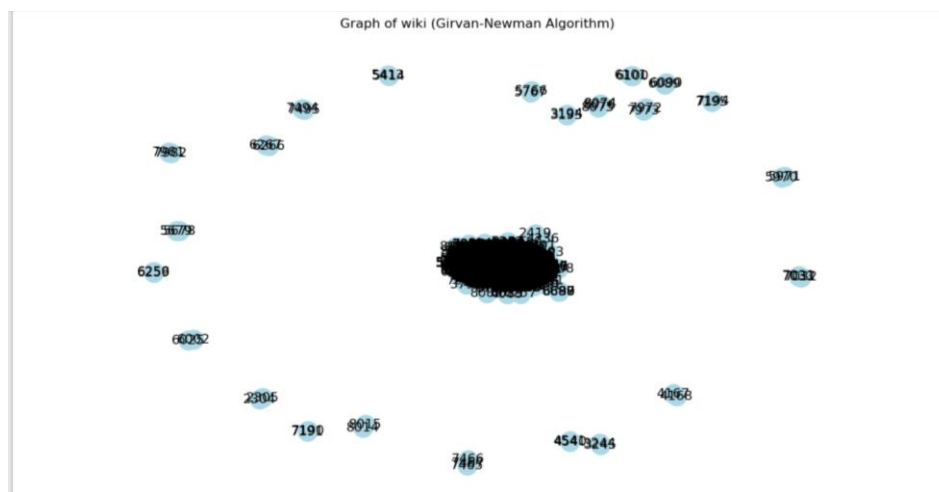
Dendrograms were used to visualise the hierarchical relatedness of the coded algorithm.

Louvain algorithm – Every node is assumed into individual community and then one node is removed from one community and shifted to another community. Change in modularity which is $\Delta q = q(\text{after}) - q(\text{before})$ is calculated and if there is positive change the iteration is kept.

Communities, time taken and modularity were the parameters involved for the determination of the better.

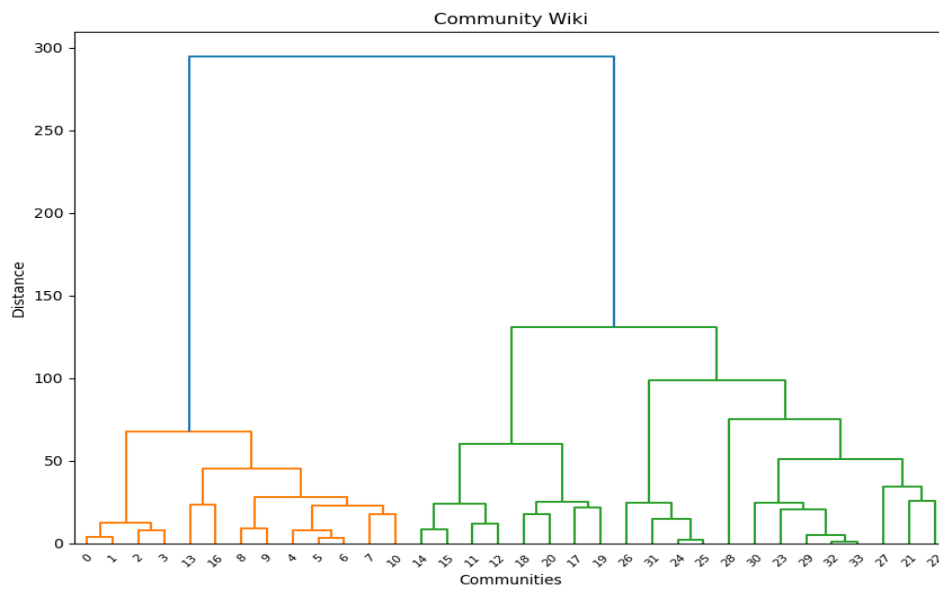
Outputs for wiki data

Ans 1) There are 24 communities

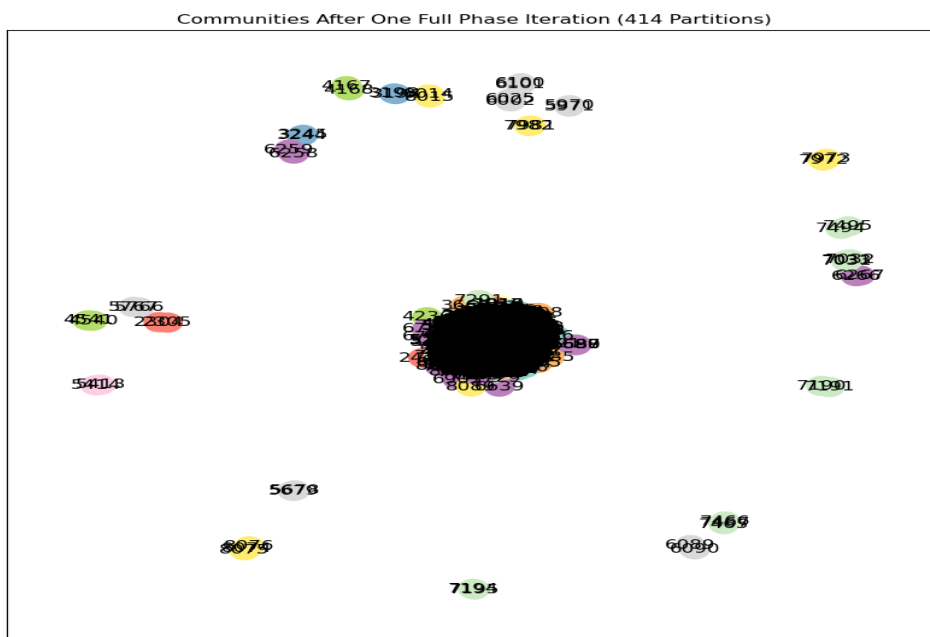


Ans 2) I looked at the plotted graphs to come with a stopping criterion for wiki dataset, the most concise one intuitively and by observing the modularity.

Ans 3) Dendrogram of girvan newman for wiki data

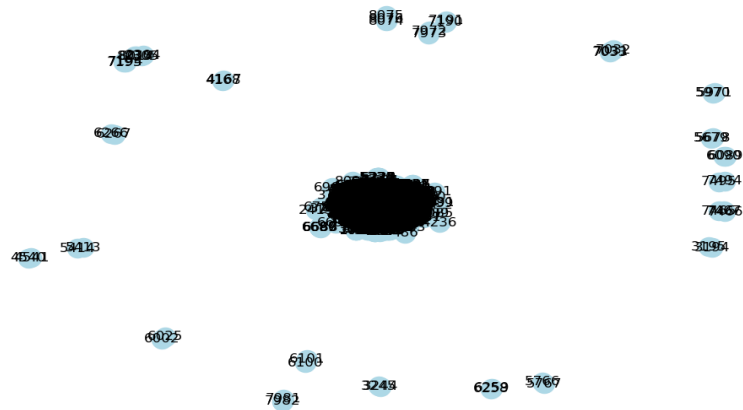


Ans 4) There are 414 partitions after one iteration of Louvain algorithm which is one less than the total total nodes.



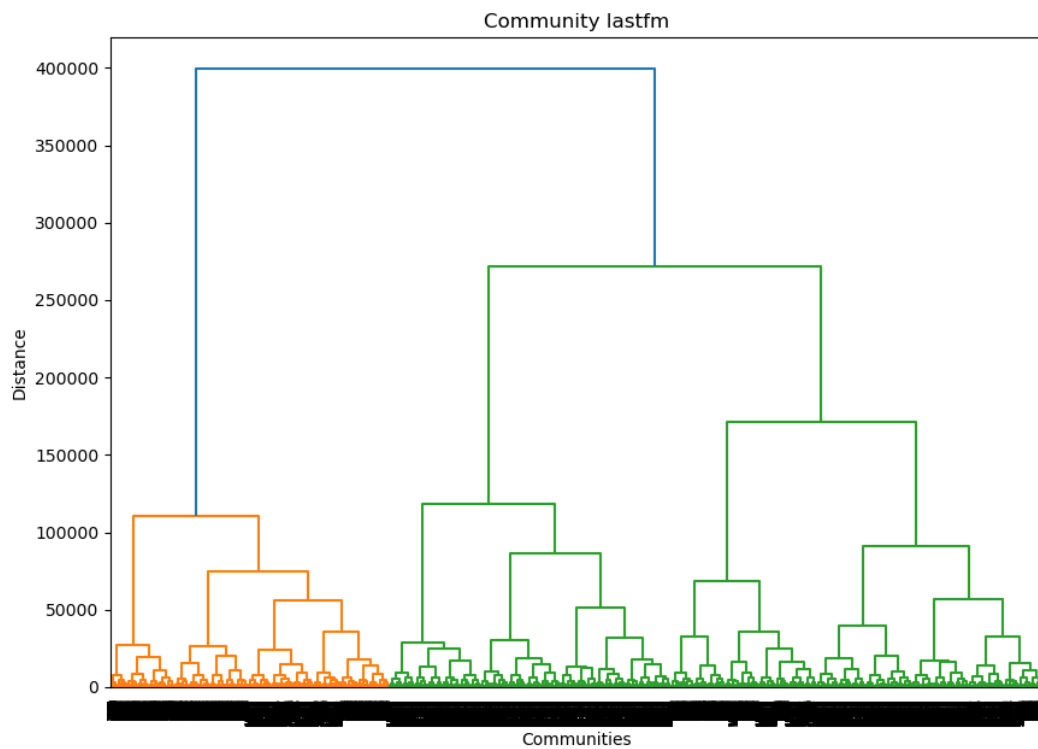
Lastfm_data

Ans 1) After many iterations of the Girvan newman algorithm the community didn't divide further and accordingly 5 communities were formed.

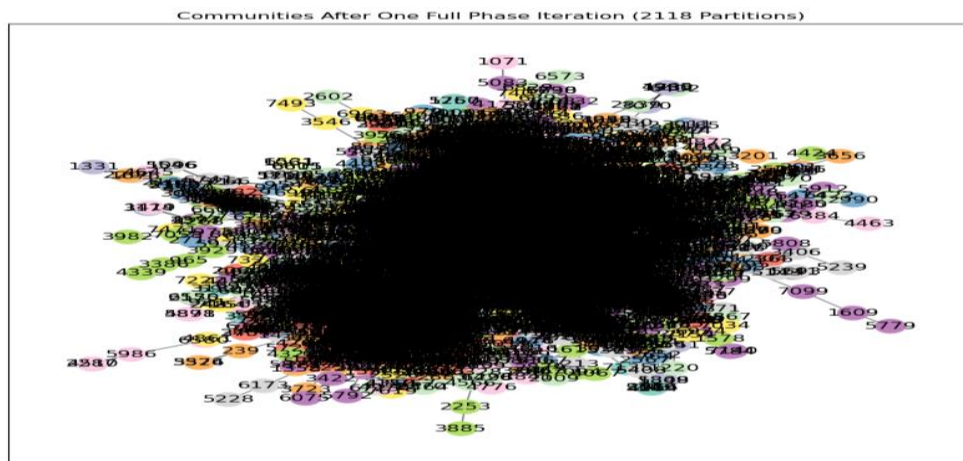


Ans 2) Checking modularity decrease is my stopping criterion.

Ans 3) Dendrogram of girvan newman on lastfm dataset.



Ans 4) There are 2118 partitions after one iteration of Louvain algorithm which is one less than the total nodes.



Q5. How would you pick the best decomposition of nodes into communities.

Ans5) I would check the modularity of the network division as it quantifies the qualitative distribution into communities by comparing the density of edges and the time taken to make communities.

6. What was the running time of the Girvan-Newman algorithm versus the Louvain algorithm on the data sets you were given?

	Girvan newman	Louvain one iteration
Wiki	Communities - 24 Modularity- 0.0051 Time — 5422 sec	Communities - 414 Time — 25 sec
Lastfm	Communities - 5 Modularity- 0.006997 Time — 1217 sec	Communities - 2118 Time — 12 sec

7. In your opinion which algorithm gave rise to better communities, and why?

Ans 7) In my opinion girvan newman gave more appropriate communities but Louvain was more time efficient. And the modularity difference was very small but the time complexity was exponentially less. I believe Louvain gave more efficient distribution of communities it took seconds to run.